

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458278.92832 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.032
Transit depth (Rp/Rs)^2	3.24 +/- 1.14 [%]
Orbital Inclination (inc)	89.82 +/- 1.65
Airmass coefficient 1 (a1)	0.999 +/- 0.038
Airmass coefficient 2 (a2)	-0.0 +/- 0.13
Scatter in the residuals of the lightcurve fit is	3.83 %
Best Comparison Star	#1 - [138, 312]
Optimal Aperture	2.96
Optimal Annulus	4.68
Transit Duration (day)	0.0922 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.032 vs 0.13031 (deviation 1.55σ)
Duration fit vs published	0.0922 d vs 0.0954 d (deviation 1.79σ)
Mid-time O–C	-1.1 min
Depth SNR	5.6
Residual scatter	3.83 %

Light curve

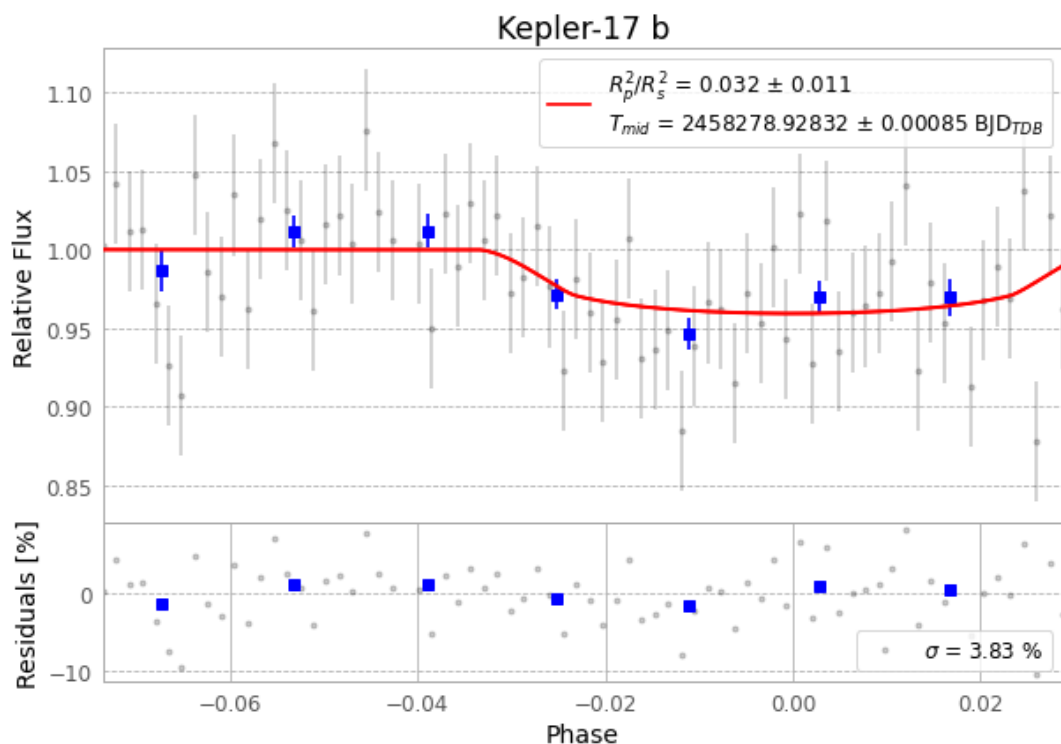


Figure 1 — FinalLightCurve_Kepler-17 b_2018-06-09.png (SHA-256 ac91ead8a96a3c30...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 249], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2880fcf706a36399...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), KEPLER-17180609073612.FITS → KEPLER-17180609111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180609.log (638ca5685e0a...), FinalLightCurve_Kepler-17 b_2018-06-09.png (ac91ead8a96a...), FinalLightCurve_Kepler-17 b_2018-06-09.csv (b8d7d5449ac9...)

Compute resources

Wall time 77.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:17Z.